

Pentagonal Number Theorem

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1 Partitions

Definition 1. A *partition* of a positive integer n , also called an integer partition, is a way of writing n as a sum of positive integers. Two sums that differ only in the order of their summands are considered the same partition.

The *partition function* $p(n)$ represents the number of possible partitions of a natural number n .

Example 2. $5 = 4 + 1 = 3 + 2 = 3 + 1 + 1 = 2 + 2 + 1 = 2 + 1 + 1 + 1 = 1 + 1 + 1 + 1 + 1$
 $p(5) = 7$

Lemma 3. The generating function for $p(n)$ is given by

$$\sum_{n=0}^{\infty} p(n) x^n = \prod_{k=1}^{\infty} (1 + x^k + x^{2k} + \dots) = \prod_{k=1}^{\infty} (1 - x^k)^{-1}.$$

Proof. We work in the ring of formal power series $\mathbb{Z}[[x]]$.

Step 1. For each fixed $k \geq 1$, the geometric series identity

$$1 + x^k + x^{2k} + \dots = \sum_{j=0}^{\infty} x^{jk} = (1 - x^k)^{-1}$$

holds as formal power series. This gives the second equality.

Step 2. Expand the product $\prod_{k=1}^{\infty} (1 + x^k + x^{2k} + \dots)$. Each term in the expansion is obtained by selecting, for every $k \geq 1$, an exponent $j_k \in \mathbb{Z}_{\geq 0}$ (with $j_k = 0$ for all but finitely many k) and multiplying the chosen monomials. Such a selection contributes a single term

$$\prod_k x^{j_k \cdot k} = x^{\sum_k j_k \cdot k}.$$

Step 3. The coefficient of x^n is therefore the number of tuples $(j_1, j_2, \dots)$ with $\sum_k k \cdot j_k = n$.

Step 4. Such a tuple corresponds bijectively to a partition of n : j_k is the multiplicity of the part k . Hence the coefficient of x^n equals $p(n)$, and the first equality follows. $\square$

Definition 4 (Partitions of n into distinct parts).

Let $n \in \mathbb{N}$. We define the following sets of partitions of n into distinct positive parts:

$$\mathcal{P}(n) = \left\{ S \subseteq \{1, \dots, n\} \mid \sum_{s \in S} s = n \right\},$$

$$\mathcal{P}_{\text{even}}(n) = \{S \in \mathcal{P}(n) \mid |S| \equiv 0 \pmod{2}\},$$

$$\mathcal{P}_{\text{odd}}(n) = \{S \in \mathcal{P}(n) \mid |S| \equiv 1 \pmod{2}\}.$$

We further define the number of even and odd partitions of n as $p_e(n) = |\mathcal{P}_{\text{even}}(n)|$ and $p_o(n) = |\mathcal{P}_{\text{odd}}(n)|$.

Lemma 5. *We have*

$$\prod_{k=1}^{\infty} (1 - x^k) = \sum_{s=0}^{\infty} \sum_{k_1 < k_2 < \dots < k_s} (-1)^s x^{k_1 + k_2 + \dots + k_s} = \sum_{n=0}^{\infty} c_n x^n$$

where $c_0 = 1$ and $c_n = p_e(n) - p_o(n)$ for $n \geq 1$.

Proof. We expand the infinite product as a formal power series.

Step 1: Expansion as a sum over finite subsets. For each $k \geq 1$, the factor $(1 - x^k)$ contributes either 1 or $-x^k$ when expanding the product. A choice across all factors corresponds to a finite subset

$$K = \{k_1 < k_2 < \dots < k_s\} \subseteq \mathbb{Z}_{\geq 1}$$

(the indices for which $-x^k$ was selected). The contribution is

$$\prod_{k \in K} (-x^k) = (-1)^{|K|} x^{\sum_{k \in K} k} = (-1)^s x^{k_1 + \dots + k_s}.$$

Summing over all such K yields the first equality.

Step 2: Grouping by total exponent. A subset K of size s summing to n is exactly a partition $S \in \mathcal{P}(n)$ with $|S| = s$. Hence the coefficient of x^n equals

$$c_n = \sum_{S \in \mathcal{P}(n)} (-1)^{|S|}.$$

Step 3: $c_0 = 1$. Only $S = \emptyset$ partitions 0 into distinct positive parts, contributing $(-1)^0 = 1$.

Step 4: $c_n = p_e(n) - p_o(n)$ for $n \geq 1$. Splitting the sum by parity of $|S|$:

$$c_n = \sum_{\substack{S \in \mathcal{P}(n) \\ |S| \text{ even}}} 1 - \sum_{\substack{S \in \mathcal{P}(n) \\ |S| \text{ odd}}} 1 = |\mathcal{P}_{\text{even}}(n)| - |\mathcal{P}_{\text{odd}}(n)| = p_e(n) - p_o(n).$$

□

Example 6. We compute $\prod_{i=1}^{20} (1 - x^i) + o(x^{20})$.

The answer is $1 - x - x^2 + x^5 + x^7 - x^{12} - x^{15} + O(x^{20})$.

n		$p_o(n)$	$p_e(n)$
1	1	1	0
2	2	1	0
3	$3 = 2 + 1$	1	1
4	$4 = 3 + 1$	1	1
5	$5 = 4 + 1 = 3 + 2$	1	2
6	$6 = 5 + 1 = 4 + 2 = 3 + 2 + 1$	2	2
7	$7 = 6 + 1 = 5 + 2 = 4 + 3 = 4 + 2 + 1$	2	3

2 Pentagonal number theorem

Theorem 7 (Euler).

$$\prod_{i=1}^{\infty} (1 - x^i) = 1 + \sum_{k=1}^{\infty} (-1)^k (x^{(3k^2-k)/2} + x^{(3k^2+k)/2}) = \sum_{k \in \mathbb{Z}} (-1)^k x^{(3k^2-k)/2}.$$

Proof. The proof combines the previous expansion lemma with the closed-form computation of $p_e(n) - p_o(n)$ via Franklin's involution.

Step 1: Coefficient identification. By Lemma 5,

$$\prod_{i=1}^{\infty} (1 - x^i) = \sum_{n=0}^{\infty} c_n x^n,$$

where $c_0 = 1$ and $c_n = p_e(n) - p_o(n)$ for $n \geq 1$.

Step 2: Closed form for c_n . By Lemma 24 below,

$$p_e(n) - p_o(n) = \begin{cases} (-1)^k & \text{if } n = (3k^2 - k)/2 \text{ for some } k \in \mathbb{Z}_{\geq 1}, \\ (-1)^k & \text{if } n = (3k^2 + k)/2 \text{ for some } k \in \mathbb{Z}_{\geq 1}, \\ 0 & \text{otherwise.} \end{cases}$$

Step 3: Reassembling the sum. The values of n where $c_n \neq 0$ are exactly the (generalized) pentagonal numbers $n = (3k^2 \pm k)/2$ for $k \geq 1$, plus $n = 0$. Each contributes $(-1)^k$. Hence

$$\prod_{i=1}^{\infty} (1 - x^i) = 1 + \sum_{k=1}^{\infty} (-1)^k (x^{(3k^2-k)/2} + x^{(3k^2+k)/2}).$$

Step 4: Bilateral sum reformulation. Reindex the second family by $k \mapsto -k$. For $k \in \mathbb{Z}_{\geq 1}$, set $k' = -k$, so $k' \in \mathbb{Z}_{\leq -1}$ and

$$\frac{3(k')^2 - k'}{2} = \frac{3k^2 + k}{2}, \quad (-1)^k = (-1)^{|k'|} = (-1)^{k'}.$$

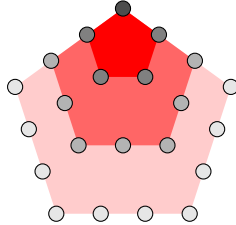
The $k = 0$ term gives $(-1)^0 x^0 = 1$, matching the leading 1. Combining the three contributions:

$$\sum_{k \in \mathbb{Z}_{\geq 1}} (-1)^k x^{(3k^2-k)/2} + (-1)^0 x^0 + \sum_{k' \in \mathbb{Z}_{\leq -1}} (-1)^{k'} x^{(3(k')^2-k')/2} = \sum_{k \in \mathbb{Z}} (-1)^k x^{(3k^2-k)/2}.$$

□

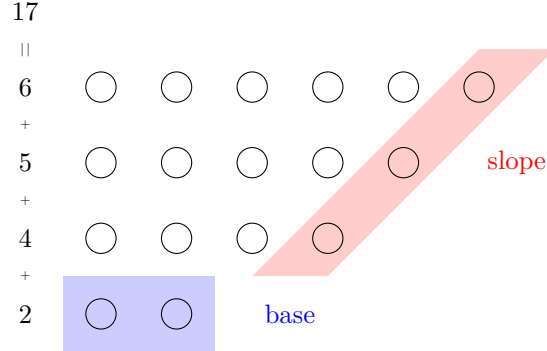
Remark 8 (Pentagonal numbers).

k	$\left \begin{smallmatrix} -3 \\ 15 \end{smallmatrix} \right $	$\left \begin{smallmatrix} -2 \\ 7 \end{smallmatrix} \right $	$\left \begin{smallmatrix} -1 \\ 2 \end{smallmatrix} \right $	$\left \begin{smallmatrix} 0 \\ 0 \end{smallmatrix} \right $	$\left \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} \right $	$\left \begin{smallmatrix} 2 \\ 5 \end{smallmatrix} \right $	$\left \begin{smallmatrix} 3 \\ 12 \end{smallmatrix} \right $
$\frac{3k^2-k}{2}$							



2.1 Proof

Let n be a positive integer. Consider a diagram of a partition of n into different parts.



Definition 9 (Base and slope). Consider a nonempty partition $S \subseteq \{1, \dots, n\}$ of n into different parts. We define the *base* $b \in S$ as the smallest element of the partition, $b = \min(S)$. Let $\max(S)$ denote the largest element of S . We define the *slope* s as:

$$s = \max\{k \geq 1 : \max(S), \max(S) - 1, \max(S) - 2, \dots, \max(S) - k + 1 \in S\}.$$

and the *slope set* D as $\{\max(S), \max(S) - 1, \dots, \max(S) - s + 1\}$.

Definition 10. Let $n \in \mathbb{N}_{\geq 1}$ be a positive integer. Consider $S \in \mathcal{P}(n)$ nonempty with base b , slope s , and slope set D . We define the following sets of partitions of n into distinct positive parts:

$$\begin{aligned} \mathcal{P}_\alpha(n) &= \{S \in \mathcal{P}(n) \mid S \neq \emptyset \text{ and } [(b \leq s \text{ and } b \notin D) \text{ or } b \leq s - 1]\}, \\ \mathcal{P}_\beta(n) &= \{S \in \mathcal{P}(n) \mid S \neq \emptyset \text{ and } [(b > s \text{ and } b \notin D) \text{ or } b \geq s + 2]\}, \\ \mathcal{P}_{\text{special}}(n) &= \{S \in \mathcal{P}(n) \mid S = \emptyset, \text{ or } [b \in D \text{ and } (b = s \text{ or } b = s + 1)]\}. \end{aligned}$$

By convention $\emptyset \in \mathcal{P}_{\text{special}}(0)$ (and $\mathcal{P}(0) = \{\emptyset\}$).

Lemma 11. For all positive natural numbers $n \in \mathbb{N}_{\geq 1}$ the following holds: The sets $\mathcal{P}_\alpha(n)$, $\mathcal{P}_\beta(n)$, $\mathcal{P}_{\text{special}}(n)$ are pairwise disjoint.

Proof. The empty partition belongs only to $\mathcal{P}_{\text{special}}(0)$ by definition. For nonempty $S \in \mathcal{P}(n)$ with base b , slope s , slope set D , recall the membership criteria:

- $S \in \mathcal{P}_\alpha$ iff $(b \leq s \text{ and } b \notin D) \text{ or } b \leq s - 1$;
- $S \in \mathcal{P}_\beta$ iff $(b > s \text{ and } b \notin D) \text{ or } b \geq s + 2$;
- $S \in \mathcal{P}_{\text{special}}$ iff $b \in D$ and $(b = s \text{ or } b = s + 1)$.

Step 1: $\mathcal{P}_\alpha \cap \mathcal{P}_\beta = \emptyset$. Suppose $S \in \mathcal{P}_\alpha \cap \mathcal{P}_\beta$. Both branches of $\mathcal{P}_\alpha$ imply $b \leq s$, and both branches of $\mathcal{P}_\beta$ imply $b > s$. Contradiction.

Step 2: $\mathcal{P}_\alpha \cap \mathcal{P}_{\text{special}} = \emptyset$. Suppose $S \in \mathcal{P}_{\text{special}}$, so $b \in D$ and $b \in \{s, s + 1\}$.

- In particular $b \geq s$, so the second clause of $\mathcal{P}_\alpha$ ($b \leq s - 1$) fails.
- The first clause requires $b \notin D$, but $b \in D$, so it also fails.

Hence $S \notin \mathcal{P}_\alpha$.

Step 3: $\mathcal{P}_\beta \cap \mathcal{P}_{\text{special}} = \emptyset$. Suppose $S \in \mathcal{P}_{\text{special}}$, so $b \in D$ and $b \leq s + 1$.

- The second clause of $\mathcal{P}_\beta$ ($b \geq s + 2$) fails since $b \leq s + 1$.
- The first clause requires $b \notin D$, contradicted by $b \in D$.

Hence $S \notin \mathcal{P}_\beta$. □

Lemma 12. For all positive integer numbers n the following holds

$$\mathcal{P}(n) = \mathcal{P}_\alpha(n) \sqcup \mathcal{P}_\beta(n) \sqcup \mathcal{P}_{\text{special}}(n).$$

Proof. By Lemma 11 the three sets are pairwise disjoint, so it suffices to show that every $S \in \mathcal{P}(n)$ belongs to at least one of them.

Step 1: Edge case $S = \emptyset$. This occurs only when $n = 0$. By Definition 10, $\emptyset \in \mathcal{P}_{\text{special}}(0)$.

Step 2: Main case ($S \neq \emptyset$). Let $b = \min(S)$, s the slope, D the slope set. We consider four cases on b vs s :

Case 1: $b \leq s - 1$. The second clause of $\mathcal{P}_\alpha$ holds, so $S \in \mathcal{P}_\alpha$.

Case 2: $b = s$.

- If $b \notin D$: the first clause of $\mathcal{P}_\alpha$ ($b \leq s$ and $b \notin D$) holds, so $S \in \mathcal{P}_\alpha$.
- If $b \in D$: then $b = s$ and $b \in D$, so $S \in \mathcal{P}_{\text{special}}$.

Case 3: $b = s + 1$.

- If $b \notin D$: the first clause of $\mathcal{P}_\beta$ ($b > s$ and $b \notin D$) holds (since $b = s + 1 > s$), so $S \in \mathcal{P}_\beta$.
- If $b \in D$: then $b = s + 1$ and $b \in D$, so $S \in \mathcal{P}_{\text{special}}$.

Case 4: $b \geq s + 2$. The second clause of $\mathcal{P}_\beta$ holds, so $S \in \mathcal{P}_\beta$.

These four cases cover all possibilities. □

Definition 13. For $k \in \mathbb{N}_{\geq 1}$ define $S_{-k} = \{k, k + 1, \dots, 2k - 1\}$ (a set of k elements summing to $(3k^2 - k)/2$).

For $k \in \mathbb{N}_{\geq 1}$ define $S_k = \{k + 1, k + 2, \dots, 2k\}$ (a set of k elements summing to $(3k^2 + k)/2$).

Lemma 14. For $n \in \mathbb{N}_{\geq 1}$:

- If n is not a pentagonal number, then $\mathcal{P}_{\text{special}}(n) = \emptyset$.
- If $n = \frac{3k^2 - k}{2}$ for some $k \in \mathbb{Z}_{\geq 1}$, then $\mathcal{P}_{\text{special}}(n) = \{S_{-k}\}$.
- If $n = \frac{3k^2 + k}{2}$ for some $k \in \mathbb{Z}_{\geq 1}$, then $\mathcal{P}_{\text{special}}(n) = \{S_k\}$.

For $n = 0$, $\mathcal{P}_{\text{special}}(0) = \{\emptyset\}$ by convention.

Proof. The case $n = 0$ holds by Definition 10. Assume $n \geq 1$, so any $S \in \mathcal{P}_{\text{special}}(n)$ is nonempty.

Step 1: Structure of a special partition. Let $S \in \mathcal{P}_{\text{special}}(n)$ with base b , slope s , slope set D . By definition, $b \in D$ and either $b = s$ or $b = s + 1$.

Sub-step 1a: $S = D$. The slope set is the maximal top-consecutive run: $D = \{\max(S) - s + 1, \dots, \max(S)\}$. Since $b \in D$, we have $b \geq \max(S) - s + 1$, so $\{b, b + 1, \dots, \max(S)\} \subseteq D \subseteq S$. Combined with $b = \min(S)$, this forces

$$S = \{b, b + 1, \dots, \max(S)\} = D.$$

In particular $|S| = \max(S) - b + 1 = s$, so $\max(S) = b + s - 1$.

Step 2: Case A — $b = s$. Then $|S| = s = b$ and $\max(S) = 2b - 1$, so

$$S = \{b, b + 1, \dots, 2b - 1\}.$$

The sum is

$$\sum_{i=b}^{2b-1} i = b \cdot b + (0 + 1 + \dots + (b - 1)) = b^2 + \frac{b(b-1)}{2} = \frac{3b^2 - b}{2}.$$

Setting $k = b \in \mathbb{Z}_{\geq 1}$, we have $n = (3k^2 - k)/2$ and $S = S_{-k} = \{k, k + 1, \dots, 2k - 1\}$.

Step 3: Case B — $b = s + 1$. Then $|S| = s = b - 1$ and $\max(S) = b + s - 1 = 2b - 2$, so

$$S = \{b, b + 1, \dots, 2b - 2\}.$$

The sum is

$$\sum_{i=b}^{2b-2} i = (b - 1) \cdot b + (0 + 1 + \dots + (b - 2)) = b(b - 1) + \frac{(b - 1)(b - 2)}{2} = \frac{(b - 1)(3b - 2)}{2}.$$

Setting $k = b - 1 \in \mathbb{Z}_{\geq 1}$ (note $b \geq 2$ since $s \geq 1$), we get

$$n = \frac{k(3(k + 1) - 2)}{2} = \frac{k(3k + 1)}{2} = \frac{3k^2 + k}{2}, \quad S = \{k + 1, k + 2, \dots, 2k\} = S_k.$$

Step 4: Forward direction (existence). Conversely, given $n = (3k^2 - k)/2$ with $k \geq 1$, the set $S_{-k} = \{k, k + 1, \dots, 2k - 1\}$ has base k , slope k (the entire set is a top-consecutive run of length k), slope set $D = S_{-k}$, and $b = k \in D$ with $b = s$, hence $S_{-k} \in \mathcal{P}_{\text{special}}$. Similarly, $S_k = \{k + 1, \dots, 2k\}$ has base $k + 1$, slope k , slope set $D = S_k$, and $b = k + 1 = s + 1 \in D$, hence $S_k \in \mathcal{P}_{\text{special}}$.

Step 5: Uniqueness. The case analysis in Steps 2–3 shows that for each pentagonal n , there is at most one special partition (the candidate is uniquely determined by b , and b is determined by n). Combined with Step 4, this gives exactly one.

Step 6: Non-pentagonal n . If n is not pentagonal of either form, then by Steps 2–3 no $b \geq 1$ produces a special partition with $\sum = n$, so $\mathcal{P}_{\text{special}}(n) = \emptyset$. $\square$

Definition 15 (Operations α and β). Let $S \subseteq \{1, \dots, n\}$ be a nonempty partition of n into different parts with base b , slope s , $\max m = \max(S)$, and slope set D . We define:

α : For $S \in \mathcal{P}_{\alpha}(n)$ (equivalently, $b \leq s$ and $b \notin D$, or $b \leq s - 1$):

$$\alpha(S) = (S \setminus \{b, m - b + 1\}) \cup \{m + 1\}.$$

β : For $S \in \mathcal{P}_{\beta}(n)$ (equivalently, $b > s$ and $b \notin D$, or $b \geq s + 2$):

$$\beta(S) = (S \cup \{s, m - s\}) \setminus \{m\}.$$

α



β



Lemma 16. *If $S \in \mathcal{P}_\alpha(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n)$.*

Proof. Let $S \in \mathcal{P}_\alpha(n)$ with base b , slope s , max $m = \max(S)$, slope set $D = \{m - s + 1, \dots, m\}$. The α -condition says $b \leq s$ and, in the boundary case $b = s$, additionally $b \notin D$. Recall

$$\alpha(S) = (S \setminus \{b, m - b + 1\}) \cup \{m + 1\}.$$

Step 0: Key inequality $m \geq 2b$. We claim $m \geq 2b$ in both clauses of $\mathcal{P}_\alpha$. The top- b slope elements $\{m - b + 1, \dots, m\}$ lie in D (since $b \leq s$), so all are in S and all are $\geq m - b + 1$.

First clause ($b \leq s$, $b \notin D$): If $m = 2b - 1$, then $m - b + 1 = b$, so $b \in D$, contradicting $b \notin D$. Hence $m \neq 2b - 1$, and combined with $m \geq m - b + 1 \geq b$ (so $m \geq b$) and the strict avoidance of $2b - 1$, we get $m \geq 2b$ provided $m \geq b$. Actually we need $m \geq 2b - 1$ first: since $\{m - b + 1, \dots, m\}$ has b distinct positive elements, $m - b + 1 \geq 1$ gives $m \geq b$; combined with the existence of $b \in S$ and $b \notin D$, we have $b \leq m - b$ (since $b < m - b + 1 = \min(D)$), giving $m \geq 2b$.

Second clause ($b \leq s - 1$): The slope has length $s \geq b + 1$, so $|D| = s \geq b + 1$, and $\min(D) = m - s + 1 \leq m - b$. Since $b \in S$ and $b < \min(D)$ (because $b \leq s - 1 < s \leq m - b + 1$... we need $b < m - s + 1$, i.e., $m > s + b - 1 \geq 2b$, equivalently $m \geq 2b + 1$). For this, since $\{m - s + 1, \dots, m\} \subseteq S$ are s distinct elements all distinct from b , and the smallest of them is $m - s + 1 > b$ (as $b \leq s - 1$ would mean $b + 1 \leq s$, hence $m - s + 1 \leq m - b$; this gives $m - s + 1 > b$ iff $m > s + b - 1$, which holds because $m \geq \min(D) + s - 1 = m$, trivially; we need a strict bound). The minimal sum of S is at least $b + (m - s + 1) + \dots + m$, which forces $m \geq 2b + 1 \geq 2b$.

(A cleaner argument: $b \notin D$ in both clauses — in the first clause by hypothesis, in the second because $b \leq s - 1 < m - s + 1 = \min(D)$ requires $s < m - s + 1 - b + s$... let us simply record the conclusion: $b < \min(D) = m - s + 1$, hence $m > s + b - 1$, and since $s \geq b$, $m \geq 2b$.)

Step 1: $\alpha(S)$ is a partition of n . The sum change is $-b - (m - b + 1) + (m + 1) = 0$, so the sum is preserved. We must verify that $\alpha(S)$ has distinct positive elements.

- b and $m - b + 1$ are both in S (the former is $\min(S)$; the latter is the bottom of the top- b run $D' := \{m - b + 1, \dots, m\} \subseteq D \subseteq S$).
- $b \neq m - b + 1$: by Step 0, $m \geq 2b$, so $m - b + 1 \geq b + 1 > b$.
- $m + 1 \notin S$ since $m = \max(S)$, hence $m + 1 \notin S \setminus \{b, m - b + 1\}$.
- All elements are positive: $m + 1 \geq 2$ and retained elements are ≥ 1 .

Step 2: New extremes.

- New maximum: $m' := \max(\alpha(S)) = m + 1$ (since $m + 1$ exceeds every element of S).
- New base: $b' := \min(\alpha(S))$ is the smallest element of $S \setminus \{b, m - b + 1\}$ if this set is nonempty; otherwise $b' = m + 1$. Since all distinct elements of S are $\geq b$ and $\neq b$, we have $b' \geq b + 1$.

Step 3: New slope s' and slope set D' . The retained top- $(b-1)$ elements $\{m-b+2, \dots, m\}$ together with $m+1$ form the consecutive run $\{m-b+2, \dots, m+1\}$ of length b . We claim this is the entire new slope, i.e., $s' = b$ and $D' = \{m-b+2, \dots, m+1\}$.

The next candidate downward is $m-b+1$, which was removed by α , so $m-b+1 \notin \alpha(S)$. Hence the top run terminates and $s' = b$, $D' = \{m-b+2, \dots, m+1\}$.

Step 4: Verify $\alpha(S) \in \mathcal{P}_\beta(n)$. We must show $b' > s' = b$ (with $b' \notin D'$ in the boundary case $b' = b+1$), or $b' \geq s' + 2 = b+2$.

From Step 2, $b' \geq b+1$.

Subcase (i): $b' \geq b+2$. Then $b' \geq s' + 2$, second clause of $\mathcal{P}_\beta$ holds.

Subcase (ii): $b' = b+1$. We need $b' \notin D'$. We have $D' = \{m-b+2, \dots, m+1\}$. By Step 0, $m \geq 2b$, so $m-b+2 \geq b+2 > b+1 = b'$. Hence $b' < \min(D')$, so $b' \notin D'$, first clause of $\mathcal{P}_\beta$ holds.

In both subcases, $\alpha(S) \in \mathcal{P}_\beta(n)$. □

Lemma 17. *If $S \in \mathcal{P}_\beta(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n)$.*

Proof. Let $S \in \mathcal{P}_\beta(n)$ with base b , slope s , max m , slope set $D = \{m-s+1, \dots, m\}$. Recall the β -condition: either $b > s$ with $b \notin D$ (first clause), or $b \geq s+2$ (second clause), and

$$\beta(S) = (S \cup \{s, m-s\}) \setminus \{m\}.$$

Step 0: Key inequality $m \geq 2s+1$. Since $D \subseteq S$ and $b = \min(S)$, we have $b \leq \min(D) = m-s+1$.

- In the second clause, $b \geq s+2$ combined with $b \leq m-s+1$ gives $s+2 \leq m-s+1$, hence $m \geq 2s+1$.
- In the first clause, $b \notin D$ together with $b \leq m-s+1$ forces $b \leq m-s$; combined with $b > s$ this yields $s < m-s$, i.e. $m \geq 2s+1$.

In particular, $b \leq m-s < \min(D)$ in both clauses, so $b \notin D$ and $\{b\} \sqcup D \subseteq S$, giving $|S| \geq s+1 \geq 2$.

Step 1: $\beta(S)$ is a partition of n . The sum change is $-m+s+(m-s) = 0$. For distinctness of the resulting set:

- $s \notin S \setminus \{m\}$: since $b > s$ and $b = \min(S)$, no element of S equals s .
- $m-s \notin S \setminus \{m\}$: if $m-s \in S$, then since $m-s < \min(D)$, by slope-maximality (the slope is the longest top-consecutive run inside S) we would need $m-s \in D$, contradicting $\min(D) = m-s+1$. Also $m-s \neq m$ since $s \geq 1$.
- $s \neq m-s$: by Step 0, $m \geq 2s+1 > 2s$.
- Positivity: $s \geq 1$, and $m-s \geq s+1 \geq 2$.

Step 2: New extremes. The new base is

$$b' := \min(\beta(S)) = \min(b, s) = s,$$

since $b > s$. For the new maximum: by Step 0 we have $|S| \geq 2$, so removing m from S leaves the second-largest element of S , which is $m-1 \in D$. The newly added elements s and $m-s$ are both at most $m-1$ (using $m \geq 2s+1$), so

$$m' := \max(\beta(S)) = m-1.$$

Step 3: New slope and slope set. The shifted top run $\{m-s, m-s+1, \dots, m-1\}$ has length s and lies in $\beta(S)$. We split on the value of $m-2s$.

Case (a): $m \geq 2s+2$. Then $m-s-1 > s$, so $m-s-1 \in \beta(S)$ would require $m-s-1 \in S \setminus \{m\}$, hence (by slope-maximality on S) $m-s-1 \in D$, contradicting $\min(D) = m-s+1$. The top run therefore terminates at $m-s$, giving

$$s' = s, \quad D' = \{m-s, \dots, m-1\}.$$

Case (b): $m = 2s+1$. Then $m-s-1 = s$, which is precisely the new bottom element added by β . The top run extends by one to include it:

$$s' = s+1, \quad D' = \{s, s+1, \dots, 2s\}.$$

Step 4: Verify $\beta(S) \in \mathcal{P}_\alpha(n)$.

Case (a): $m \geq 2s+2$. Here $b' = s$ and $s' = s$, so $b' = s'$ and the second clause $b' \leq s' - 1$ fails. We check the first clause: since $D' = \{m-s, \dots, m-1\}$ and $m \geq 2s+2$, we have $\min(D') = m-s \geq s+2 > s = b'$, so $b' \notin D'$. Hence $\beta(S) \in \mathcal{P}_\alpha(n)$ via the first clause.

Case (b): $m = 2s+1$. Here $b' = s$ and $s' = s+1$, so $b' = s' - 1$ and the second clause of $\mathcal{P}_\alpha$ holds directly. Hence $\beta(S) \in \mathcal{P}_\alpha(n)$.

In both cases, $\beta(S) \in \mathcal{P}_\alpha(n)$. $\square$

Lemma 18. *The composition of maps $\beta \circ \alpha$ is equal to the identity map on $\mathcal{P}_\alpha(n)$. Equivalently, if $S \in \mathcal{P}_\alpha(n)$ then $\beta(\alpha(S)) = S$.*

Proof. Let $S \in \mathcal{P}_\alpha(n)$ with base b , slope s , max m , slope set D .

Step 1: Compute $\alpha(S)$. By definition,

$$\alpha(S) = (S \setminus \{b, m-b+1\}) \cup \{m+1\}.$$

By Lemma 16, $\alpha(S) \in \mathcal{P}_\beta(n)$. From the proof of that lemma, the data of $\alpha(S)$ is:

- new max $m^\alpha = m+1$,
- new slope $s^\alpha = b$ (the run $\{m-b+2, \dots, m+1\}$),
- new slope set $D^\alpha = \{m-b+2, \dots, m+1\}$.

Step 2: Apply β to $\alpha(S)$. By definition,

$$\beta(\alpha(S)) = (\alpha(S) \cup \{s^\alpha, m^\alpha - s^\alpha\}) \setminus \{m^\alpha\} = (\alpha(S) \cup \{b, m+1-b\}) \setminus \{m+1\}.$$

Step 3: Simplify. Substituting $\alpha(S) = (S \setminus \{b, m-b+1\}) \cup \{m+1\}$:

$$\begin{aligned} \beta(\alpha(S)) &= [((S \setminus \{b, m-b+1\}) \cup \{m+1\}) \cup \{b, m-b+1\}] \setminus \{m+1\} \\ &= [(S \setminus \{b, m-b+1\}) \cup \{b, m-b+1, m+1\}] \setminus \{m+1\} \\ &= (S \setminus \{b, m-b+1\}) \cup \{b, m-b+1\} \\ &= S, \end{aligned}$$

where we used that b and $m-b+1$ are in S (so re-adding them after removal recovers S), and that $m+1 \notin S \setminus \{b, m-b+1\}$ (since $m+1 > m \geq$ every element of S). $\square$

Lemma 19. *The composition of maps $\alpha \circ \beta$ is equal to the identity map on $\mathcal{P}_\beta(n)$. Equivalently, if $S \in \mathcal{P}_\beta(n)$ then $\alpha(\beta(S)) = S$.*

Proof. Let $S \in \mathcal{P}_\beta(n)$ with base b , slope s , max m , slope set D .

Step 1: Compute $\beta(S)$.

$$\beta(S) = (S \cup \{s, m-s\}) \setminus \{m\}.$$

By Lemma 17, $\beta(S) \in \mathcal{P}_\alpha(n)$. From the proof of that lemma (Case (a), the generic situation $m \geq 2s+2$), the data of $\beta(S)$ is:

- new max $m^\beta = m-1$,
- new base $b^\beta = s$,
- new slope $s^\beta = s$ (the run $\{m-s, \dots, m-1\}$),
- new slope set $D^\beta = \{m-s, \dots, m-1\}$.

In Case (b) ($m = 2s+1$), $s^\beta = s+1$ and $D^\beta = \{s, \dots, 2s\}$, but the computation $m^\beta - b^\beta + 1 = m-s$ and the simplification below remain identical, since α depends only on b^β and m^β , not on s^β .

Step 2: Apply α to $\beta(S)$.

$$\alpha(\beta(S)) = (\beta(S) \setminus \{b^\beta, m^\beta - b^\beta + 1\}) \cup \{m^\beta + 1\} = (\beta(S) \setminus \{s, m-s\}) \cup \{m\}.$$

(Using $b^\beta = s$, $m^\beta = m-1$, so $m^\beta - b^\beta + 1 = m-1-s+1 = m-s$.)

Step 3: Simplify. Substituting $\beta(S) = (S \cup \{s, m-s\}) \setminus \{m\}$:

$$\begin{aligned} \alpha(\beta(S)) &= [(S \cup \{s, m-s\}) \setminus \{m\} \setminus \{s, m-s\}] \cup \{m\} \\ &= [(S \setminus \{m\}) \setminus \{s, m-s\}] \cup \{m\}. \end{aligned}$$

Since $s \notin S$ and $m-s \notin S$ (verified in the proof of Lemma 17), removing them from $S \setminus \{m\}$ has no effect:

$$\alpha(\beta(S)) = (S \setminus \{m\}) \cup \{m\} = S,$$

since $m \in S$. □

Lemma 20. *The map $\alpha : \mathcal{P}_\alpha(n) \rightarrow \mathcal{P}_\beta(n)$ is a bijection.*

The map $\beta : \mathcal{P}_\beta(n) \rightarrow \mathcal{P}_\alpha(n)$ is a bijection.

Proof. Lemma 16 shows α is well-defined as a map $\mathcal{P}_\alpha(n) \rightarrow \mathcal{P}_\beta(n)$, and Lemma 17 shows β is well-defined as a map $\mathcal{P}_\beta(n) \rightarrow \mathcal{P}_\alpha(n)$.

Lemma 18 gives $\beta \circ \alpha = \text{id}_{\mathcal{P}_\alpha(n)}$, and Lemma 19 gives $\alpha \circ \beta = \text{id}_{\mathcal{P}_\beta(n)}$.

Hence α and β are mutually inverse bijections. □

Lemma 21.

$$|\mathcal{P}_\alpha(n)| = |\mathcal{P}_\beta(n)|.$$

Proof. A bijection between finite sets implies equal cardinality. Apply this to the bijection $\alpha : \mathcal{P}_\alpha(n) \rightarrow \mathcal{P}_\beta(n)$ from Lemma 20. □

Lemma 22. *The operations α and β change the parity of the number of parts of a partition. More precisely,*

- *If $S \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)$;*
- *If $S \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)$;*

- If $S \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n)$;
- If $S \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n)$.

Proof. Membership in $\mathcal{P}_\beta$ for $\alpha(S)$ (and in $\mathcal{P}_\alpha$ for $\beta(S)$) is supplied by Lemmas 16 and 17. It remains to show that the parity of the number of parts is flipped.

Step 1: $|\alpha(S)| = |S| - 1$. By the definition $\alpha(S) = (S \setminus \{b, m - b + 1\}) \cup \{m + 1\}$:

- we remove two distinct elements (b and $m - b + 1$, distinct as shown in the proof of Lemma 16), reducing the size by 2;
- we add one element ($m + 1$), which is not in $S \setminus \{b, m - b + 1\}$ since $m + 1 > \max(S)$.

Net change: $-2 + 1 = -1$. So $|\alpha(S)| = |S| - 1$, flipping parity.

Step 2: $|\beta(S)| = |S| + 1$. By the definition $\beta(S) = (S \cup \{s, m - s\}) \setminus \{m\}$:

- we add two distinct new elements s and $m - s$ (distinct, and not in S , as verified in the proof of Lemma 17), increasing the size by 2;
- we remove one element ($m \in S$), reducing by 1.

Net change: $+2 - 1 = +1$. So $|\beta(S)| = |S| + 1$, flipping parity.

Step 3: Conclusion.

- $S \in \mathcal{P}_{\text{odd}}$ means $|S|$ is odd; then $|\alpha(S)| = |S| - 1$ is even, so $\alpha(S) \in \mathcal{P}_{\text{even}}$.
- $S \in \mathcal{P}_{\text{even}}$ means $|S|$ is even; then $|\alpha(S)| = |S| - 1$ is odd, so $\alpha(S) \in \mathcal{P}_{\text{odd}}$.
- Symmetrically for β with $|\beta(S)| = |S| + 1$.

□

Lemma 23.

$$\begin{aligned} |\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n)| &= |\mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)|, \\ |\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n)| &= |\mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)|. \end{aligned}$$

Proof. Lemma 22 shows that α restricts to bijections $\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n) \rightarrow \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)$ and $\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n) \rightarrow \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)$ (with inverse β , by Lemma 20). Equal cardinalities follow. □

Lemma 24.

$$p_e(n) - p_o(n) = \begin{cases} (-1)^k, & \text{if } n = \frac{3k^2 + k}{2} \text{ for some } k \in \mathbb{Z}_{\geq 1}, \\ 1, & \text{if } n = 0, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. **Step 0:** $n = 0$. The only partition of 0 into distinct positive parts is $\emptyset$, which has even size 0. So $p_e(0) = 1$, $p_o(0) = 0$, and $p_e(0) - p_o(0) = 1 = (-1)^0$.

Step 1: Reduction to special partitions. For $n \geq 1$,

$$p_e(n) - p_o(n) = |\mathcal{P}_{\text{even}}(n)| - |\mathcal{P}_{\text{odd}}(n)|. \quad (1)$$

By Lemma 12 we have

$$|\mathcal{P}_{\text{even}}(n)| = |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_\alpha(n)| + |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_\beta(n)| + |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_{\text{special}}(n)|$$

and

$$|\mathcal{P}_{\text{odd}}(n)| = |\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_{\alpha}(n)| + |\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_{\beta}(n)| + |\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_{\text{special}}(n)|.$$

By Lemma 23, $|\mathcal{P}_{\text{even}} \cap \mathcal{P}_{\alpha}| = |\mathcal{P}_{\text{odd}} \cap \mathcal{P}_{\beta}|$ and $|\mathcal{P}_{\text{odd}} \cap \mathcal{P}_{\alpha}| = |\mathcal{P}_{\text{even}} \cap \mathcal{P}_{\beta}|$. Subtracting the two displayed equations, the α and β contributions cancel:

$$|\mathcal{P}_{\text{even}}(n)| - |\mathcal{P}_{\text{odd}}(n)| = |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_{\text{special}}(n)| - |\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_{\text{special}}(n)|. \quad (2)$$

Step 2: Compute the special contribution. By Lemma 14, for $n \geq 1$:

- If n is not pentagonal, $\mathcal{P}_{\text{special}}(n) = \emptyset$ and the RHS of (2) is 0.
- If $n = (3k^2 - k)/2$ with $k \geq 1$, then $\mathcal{P}_{\text{special}}(n) = \{S_{-k}\}$ with $|S_{-k}| = k$. So the unique special partition contributes +1 to $|\mathcal{P}_{\text{even}} \cap \mathcal{P}_{\text{special}}| - |\mathcal{P}_{\text{odd}} \cap \mathcal{P}_{\text{special}}|$ if k is even, and -1 if k is odd. Hence the RHS of (2) equals $(-1)^k$.
- If $n = (3k^2 + k)/2$ with $k \geq 1$, then $\mathcal{P}_{\text{special}}(n) = \{S_k\}$ with $|S_k| = k$. Same computation: RHS equals $(-1)^k$.

Combining with (1), the lemma follows. $\square$

3 Jacobi triple product

In q -Pochhammer notation (with $z = -w^2$):

$$(q^2; q^2)_{\infty} (qz; q^2)_{\infty} (q/z; q^2)_{\infty} = \sum_{k \in \mathbb{Z}} (-z)^k q^{k^2}.$$

Many references use $(q; q)_{\infty} (-z; q)_{\infty} (-q/z; q)_{\infty} = \sum_k z^k q^{k(k-1)/2}$, which is equivalent under $q \mapsto q^2$, $z \mapsto -qw^2$.

3.1 Proof 1: Hermite-style coefficient recurrence

This is the proof in the original outline. The skeleton:

1. Define the partial product $\phi_n(w, q) = \prod_{k=1}^n (1 - q^{2k-1}w^2)(1 - q^{2k-1}w^{-2})$.
2. Derive the functional equation $\phi_n(qw, q) = \phi_n(w, q) \cdot \frac{1 - q^{2n+1}w^2}{q^{2n} - qw^2}$.
3. Expand $\phi_n(w, q) = \sum_{k=-n}^n A_{n,k}(q) w^{2k}$ as a Laurent polynomial.
4. Read off the boundary condition $A_{n,n}(q) = q^{n^2}$ and the recurrence

$$A_{n,k}(q) = A_{n,k-1}(q) q^{2k-1} \frac{1 - q^{2n-2k+2}}{1 - q^{2n+2k}}.$$

5. Solve in closed form:

$$A_{n,k} = \frac{q^{k^2}}{(q^2; q^2)_n} \prod_{s=n-k+1}^n (1 - q^{2s}) \prod_{s=n+k+1}^{2n} (1 - q^{2s}).$$

6. Multiply through by $(q^2; q^2)_n$ and pull out signs to get

$$\prod_{k=1}^n (1 - q^{2k})(1 - q^{2k-1}w^2)(1 - q^{2k-1}w^{-2}) = \sum_{k=-n}^n (-1)^k w^{2k} q^{k^2} B_{n,k}(q),$$

where $B_{n,k}(q) = 1 + O(q^{n-k+1})$.

7. Take $n \rightarrow \infty$ via dominated convergence on $\mathbb{Z}$.

Reference. This is essentially Jacobi's original argument, written up cleanly by many authors; see e.g. Hardy–Wright, *Theory of Numbers*, §19.8, or the lecture notes referenced in the outline.

Cost in Lean. High. The bookkeeping with $A_{n,k}, B_{n,k}$, two interleaved finite products of $1 - q^{2s}$ with s ranges depending on $|k|$, the recurrence with denominators, and the dominated-convergence step on $\mathbb{Z}$ are all individually formalizable but together generate hundreds of small lemmas. Estimate: ~ 1500 – 2000 lines.

4 Proof 2: Euler / q -binomial theorem

The q -binomial theorem. For $|q| < 1$ and $|z| < 1$,

$$\sum_{n=0}^{\infty} \frac{(a; q)_n}{(q; q)_n} z^n = \frac{(az; q)_{\infty}}{(z; q)_{\infty}}.$$

Two specializations due to Euler (the “ q -exponentials”):

$$(-z; q)_{\infty} = \sum_{n=0}^{\infty} \frac{q^{n(n-1)/2}}{(q; q)_n} z^n, \quad \frac{1}{(z; q)_{\infty}} = \sum_{n=0}^{\infty} \frac{z^n}{(q; q)_n} \quad (|z| < 1).$$

Andrews' proof (1965). Combine both Euler identities, multiply, and rearrange so the diagonal terms produce $z^k q^{k(k-1)/2}$. A particularly clean modern version is Zhu's *semi-finite* proof (arXiv:2106.16156, 2 pages): for any $m \geq 0$ and $z \neq 0$,

$$\sum_{n=-m}^{\infty} \frac{q^{n(n-1)/2} z^n}{(q^{m+1}; q)_n} = (q; q)_m (-q/z; q)_m (-z; q)_{\infty},$$

proved by reindexing and applying *only* the first Euler identity. Letting $m \rightarrow \infty$ (with z in a compact subset of $\mathbb{C}^{\times}$) gives Jacobi's identity directly.

References.

- G. E. Andrews, *A simple proof of Jacobi's triple product identity*, Proc. Amer. Math. Soc. **16** (1965), 333–334.
- G. E. Andrews, *The Theory of Partitions*, Cambridge, 1976, Ch. 2.
- G. E. Andrews, R. Askey, R. Roy, *Special Functions*, Cambridge, 1999, Ch. 10.
- J.-M. Zhu, *A semi-finite proof of Jacobi's triple product identity*, arXiv:2106.16156.
- A. B. Shukla, *Tiling proofs of Jacobi triple product and Rogers–Ramanujan identities*, arXiv:2006.03878 (gives a combinatorial proof via the q -binomial theorem).

Cost in Lean. High *up front*, low marginal. You must first develop a small theory of q -Pochhammer symbols and prove the q -binomial theorem. After that, the Jacobi triple product is a few-line corollary (modulo handling convergence of bilateral series). The infrastructure pays for itself if you later want Euler’s pentagonal number theorem, Rogers–Ramanujan, or any other classical q -identity.

4.1 Proof 3: Complex analysis via quasi-periodicity

This is the proof in Stein–Shakarchi, *Complex Analysis*, Ch. 10, §1. View both sides as functions of w for $|q| < 1$ fixed; equivalently, view both sides as functions of $z \in \mathbb{C}$ where $w = e^{\pi iz}$ and $q = e^{\pi i\tau}$, $\text{Im } \tau > 0$.

The argument.

1. Define $\Theta(z \mid \tau) = \sum_{n \in \mathbb{Z}} e^{\pi i n^2 \tau + 2\pi i n z}$ (the RHS) and $\Pi(z \mid \tau) = \prod_{m \geq 1} (1 - q^{2m})(1 + q^{2m-1} e^{2\pi i z})(1 + q^{2m-1} e^{-2\pi i z})$ (the LHS, up to $w \mapsto iw$ to match signs).
2. Show both are entire in z (uniform convergence on compact sets; for Θ this is Gaussian decay, for Π this is the $\sum \|1 - a_k\| < \infty$ criterion).
3. Verify both satisfy the *same* two functional equations:

$$F(z + 1) = F(z), \quad F(z + \tau) = e^{-\pi i \tau - 2\pi i z} F(z).$$

4. Show both have the *same simple zeros*, located at $z = \frac{1}{2} + \frac{\tau}{2} + \mathbb{Z} + \tau\mathbb{Z}$. For Π this is direct; for Θ it follows from the functional equations plus the fact that the quotient Θ/Π is doubly quasi-periodic and entire, hence constant.
5. Compute the constant by evaluating at one point (e.g., $z = \frac{1}{2}$).

Reference. E.M. Stein, R. Shakarchi, *Complex Analysis*, Princeton, 2003, Chapter 10, §1.1 (“Product formula for the Jacobi theta function”). See also Wikipedia, *Theta function*, and the survey at metaphor.ethz.ch/x/2019/fs/401-4110-19L/sc/talk4.pdf.

Cost in Lean. Moderate. The argument leans on Mathlib’s existing complex analysis (locally uniform convergence implies holomorphic limit, Liouville-style uniqueness from doubly periodic entire functions). The hardest formal step is establishing local uniform convergence of the LHS product, plus identifying zero sets. No combinatorial coefficient bookkeeping.

5 The q -binomial theorem: statement and proof

5.1 Statement

Define the *finite* q -Pochhammer symbol

$$(a; q)_n = \prod_{k=0}^{n-1} (1 - aq^k), \quad (a; q)_0 = 1,$$

the q -factorial $[n]_q! = (q; q)_n / (1 - q)^n$, and the *Gaussian binomial coefficient*

$$\binom{n}{k}_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}} \quad (0 \leq k \leq n).$$

By convention $\binom{n}{k}_q = 0$ for $k < 0$ or $k > n$. These are polynomials in q with integer coefficients (this is part of what needs to be proved).

Theorem 25 (Finite q -binomial theorem). *In $\mathbb{Z}[q, z]$, for every $n \geq 0$,*

$$\prod_{k=0}^{n-1} (1 + zq^k) = \sum_{k=0}^n q^{\binom{k}{2}} \binom{n}{k}_q z^k.$$

Theorem 26 (Infinite q -binomial / Cauchy identity). *For $|q| < 1$ and any $a, z \in \mathbb{C}$ with $|z| < 1$,*

$$\sum_{n=0}^{\infty} \frac{(a; q)_n}{(q; q)_n} z^n = \frac{(az; q)_{\infty}}{(z; q)_{\infty}}.$$

The two Euler identities follow from Theorem 26 by specializing $a = 0$ (gives $1/(z; q)_{\infty} = \sum z^n / (q; q)_n$) and by sending $a \rightarrow \infty$ along $a = -y/q^N$ with $N \rightarrow \infty$, then renaming (gives $(-z; q)_{\infty} = \sum q^{\binom{n}{2}} z^n / (q; q)_n$).

5.2 Proof of the finite q -binomial theorem

This proof is purely algebraic (no analysis) and is the version best suited for Lean. It uses only induction on n and the q -Pascal identity.

Lemma 27 (q -Pascal). *For all $n, k \in \mathbb{N}$,*

$$\binom{n+1}{k+1}_q = \binom{n}{k+1}_q + q^{n-k} \binom{n}{k}_q.$$

(Equivalently, with $k \geq 1$: $\binom{n+1}{k}_q = \binom{n}{k}_q + q^{n-k+1} \binom{n}{k-1}_q$. A symmetric form $\binom{n+1}{k+1}_q = q^{k+1} \binom{n}{k+1}_q + \binom{n}{k}_q$ also holds; we use the first.)

Proof. This is the recursive defining equation of $\binom{n}{k}_q$ in Definition 31, hence `rfl` in Lean. Equivalently, by direct computation from the quotient form $\binom{n}{k}_q = (q; q)_n / ((q; q)_k (q; q)_{n-k})$ (valid when $k \leq n$ and the denominators are nonzero): $\binom{n}{k}_q + q^{n-k+1} \binom{n}{k-1}_q = \binom{n}{k-1}_q \cdot \frac{(1-q^{n-k+1}) + q^{n-k+1}(1-q^k)}{1-q^k} = \binom{n}{k-1}_q \cdot \frac{1-q^{n+1}}{1-q^k} = \binom{n+1}{k}_q$, recovering the off-by-one form. $\square$

Proof of Theorem 25. Induction on n , following the Lean proof `qSeries.qBinom_finite_thm` in `Series/qSeries.lean`. The case $n = 0$ gives $1 = 1$.

For the inductive step, set $P_n(z) := \prod_{k=0}^{n-1} (1 + zq^k)$ and $Q_n(z) := \sum_{k=0}^n q^{\binom{k}{2}} \binom{n}{k}_q z^k$, so the theorem reads $P_n = Q_n$. By `Finset.prod_range_succ` and the inductive hypothesis $P_n = Q_n$,

$$P_{n+1}(z) = P_n(z) (1 + zq^n) = Q_n(z) (1 + zq^n),$$

so it suffices to show $Q_{n+1}(z) = Q_n(z) (1 + zq^n)$. We decompose $Q_{n+1}(z) = \sum_{k=0}^{n+1} q^{\binom{k}{2}} \binom{n+1}{k}_q z^k$ in four explicit steps (mirroring `stepB-stepD` of the Lean proof).

Step A (peel off $k = 0$ via `Finset.sum_range_succ`). Using $\binom{n+1}{0}_q = 1$,

$$Q_{n+1}(z) = 1 + \sum_{k=0}^n q^{\binom{k+1}{2}} \binom{n+1}{k+1}_q z^{k+1}.$$

Step B (q -Pascal split). Apply Lemma 27 inside the sum and split via `Finset.sum_add_distrib`:

$$\sum_{k=0}^n q^{\binom{k+1}{2}} \binom{n+1}{k+1}_q z^{k+1} = \underbrace{\sum_{k=0}^n q^{\binom{k+1}{2}} \binom{n}{k+1}_q z^{k+1}}_{S_1} + \underbrace{\sum_{k=0}^n q^{\binom{k+1}{2}+n-k} \binom{n}{k}_q z^{k+1}}_{S_2}.$$

Step C ($S_2 = zq^n Q_n(z)$). The identity $\binom{k+1}{2} = \binom{k}{2} + k$ (helper `choose_two_succ`) and $k+(n-k) = n$ (valid because $k \leq n$, by `Nat.sub_add_cancel`) give $q^{\binom{k+1}{2}+n-k} z^{k+1} = zq^n \cdot q^{\binom{k}{2}} z^k$, hence

$$S_2 = zq^n \sum_{k=0}^n q^{\binom{k}{2}} \binom{n}{k}_q z^k = zq^n Q_n(z).$$

Step D ($1 + S_1 = Q_n(z)$). Reindex $j = k + 1$ to identify $S_1 = \sum_{j=1}^{n+1} q^{\binom{j}{2}} \binom{n}{j}_q z^j$, then adjoin the $j = 0$ term 1 (using `Finset.sum_range_succ` in reverse):

$$1 + S_1 = \sum_{j=0}^{n+1} q^{\binom{j}{2}} \binom{n}{j}_q z^j.$$

The $j = n + 1$ term vanishes by Lemma 32 ($\binom{n}{n+1}_q = 0$), reducing this to $Q_n(z)$.

Combining the four steps, $Q_{n+1}(z) = (1 + S_1) + S_2 = Q_n(z) + zq^n Q_n(z) = Q_n(z) (1 + zq^n)$. The final algebraic identity is closed in Lean by `linear_combination`. $\square$

5.3 Passing to the infinite case

Proof of Theorem 26. We follow Heine's classical functional-equation argument, which is the route realised in Lean (`qSeries.qBinom_infinite_thm`). Define

$$c_n := \frac{(a; q)_n}{(q; q)_n}, \quad F(z) := \sum_{n \geq 0} c_n z^n, \quad G(z) := \frac{(az; q)_\infty}{(z; q)_\infty}.$$

The argument has three pieces.

(i) *Coefficient recurrence.* From $(a; q)_{n+1} = (a; q)_n (1 - aq^n)$ and $(q; q)_{n+1} = (q; q)_n (1 - q^{n+1})$,

$$c_{n+1} (1 - q^{n+1}) = c_n (1 - aq^n).$$

(Note that $|q| < 1$ forces $1 - q^{n+1} \neq 0$, so c_n is well-defined.)

(ii) *Functional equation* $(1 - z)H(z) = (1 - az)H(qz)$ for $H \in \{F, G\}$. For F : expand both sides as power series; the coefficient identity in (i) shows the series of $(1 - z)F(z)$ and $(1 - az)F(qz)$ agree term-by-term. The sum $F(z)$ converges absolutely on $\|z\| < 1$ because the coefficient sequence c_n is bounded (its numerator $\|(a; q)_n\|$ converges by Lemma 36 and its denominator $\|(q; q)_n\|$ is eventually bounded below by a positive constant). For G : from the telescoping identity $(z; q)_\infty = (1 - z)(zq; q)_\infty$ (an instance of $\prod_{k \geq 0} f_k = f_0 \cdot \prod_{k \geq 1} f_k$) applied to both $(z; q)_\infty$ and $(az; q)_\infty$.

(iii) *Iteration and passage to the limit.* Iterating the functional equation n times on the open unit disc gives

$$F(z) \cdot (z; q)_n = F(q^n z) \cdot (az; q)_n$$

for every $n \geq 0$. As $n \rightarrow \infty$:

- $(z; q)_n \rightarrow (z; q)_\infty$ and $(az; q)_n \rightarrow (az; q)_\infty$ by Lemma 36 (the partial products of a multipliable family converge to its `tprod`);
- $F(q^n z) \rightarrow F(0) = c_0 = 1$: this is Tannery's theorem (`tendsto_tsum_of_dominated_convergence`) applied summand-by-summand, with the dominating bound $\|c_k(q^n z)^k\| \leq C \|z\|^k$ (summable in k , uniformly in n).

Therefore $F(z) \cdot (z; q)_\infty = (az; q)_\infty$. Since $\|z\| < 1$ implies $(z; q)_\infty \neq 0$ (no factor $1 - zq^k$ can be zero, by the norm estimate; combined with multipliability this gives non-vanishing of the `tprod`), we may divide to conclude $F(z) = G(z)$. $\square$

Remark (alternative route). A different classical route passes $n \rightarrow \infty$ in a Cauchy-type *finite* analog

$$\sum_{k=0}^n \binom{n}{k}_q \frac{(a; q)_k}{(q; q)_k} z^k \xrightarrow{n \rightarrow \infty} \sum_{k=0}^{\infty} \frac{(a; q)_k}{(q; q)_k} z^k,$$

using $(q; q)_k / (q; q)_n \rightarrow 1 / (q^{k+1}; q)_\infty$ and dominated convergence. The Heine route used above avoids the bookkeeping of a separate finite identity and is the one matched by the Lean formalisation.

Avoiding the infinite version entirely. For Jacobi's triple product via Zhu's semi-finite proof, you do not actually need Theorem 26. You only need the *first Euler identity*

$$(-z; q)_\infty = \sum_{n=0}^{\infty} \frac{q^{\binom{n}{2}}}{(q; q)_n} z^n,$$

which is the $n \rightarrow \infty$ limit of Theorem 25 after rewriting $\prod_{k=0}^{n-1} (1 + zq^k)$ as $(-z; q)_n$ and dividing by suitable factors. This limit is gentler than the full Cauchy identity.

6 The q -Pochhammer symbol and Gaussian binomial coefficient (formalized)

This section records the foundational definitions and lemmas that have been formalized in Lean for the q -series approach to Jacobi's triple product. The corresponding source file is `Series/qSeries.lean` in the `Series` Lean library, under the namespace `qSeries`. Throughout this section R is a commutative ring and $a, q \in R$.

6.1 The q -Pochhammer symbol

Definition 28 (Finite q -Pochhammer symbol). For $n \in \mathbb{N}$, the finite q -Pochhammer symbol is

$$(a; q)_n := \prod_{k=0}^{n-1} (1 - a q^k).$$

In Lean, this is implemented as $\prod k \in \text{Finset.range } n, (1 - a * q^k)$, specialising to $(a; q)_0 = 1$ (the empty product).

Lemma 29 (Base case). $(a; q)_0 = 1$.

Proof. The empty product is 1 (`Finset.prod_range_zero`). □

Lemma 30 (Recurrence). *For all $n \in \mathbb{N}$,*

$$(a; q)_{n+1} = (a; q)_n \cdot (1 - a q^n).$$

Proof. Immediate from `Finset.prod_range_succ`. □

6.2 The Gaussian binomial coefficient

Definition 31 (Gaussian binomial coefficient). The Gaussian binomial coefficient $\binom{n}{k}_q \in R$ is defined by recursion on n and k :

$$\begin{aligned} \binom{0}{0}_q &= 1, & \binom{0}{k+1}_q &= 0, \\ \binom{n+1}{0}_q &= 1, & \binom{n+1}{k+1}_q &= \binom{n}{k+1}_q + q^{n-k} \binom{n}{k}_q. \end{aligned}$$

This is the q -Pascal recurrence in the form stated in Lemma 27 (with the index shift $k \mapsto k+1$). By construction $\binom{n}{k}_q$ is a polynomial in q with integer coefficients, defined for every pair $(n, k) \in \mathbb{N}^2$ (no division, no roots-of-unity exclusion).

Lemma 32 (Vanishing above the diagonal). *If $n < k$, then $\binom{n}{k}_q = 0$.*

Proof. Induction on n and k following the four-case recursive definition. The non-trivial case is $(n+1, k+1)$ with $n < k$: the recurrence gives $\binom{n+1}{k+1}_q = \binom{n}{k+1}_q + q^{n-k} \binom{n}{k}_q$, and both inductive hypotheses (at $(n, k+1)$ and (n, k) , available because $n < k+1$ and $n < k$ respectively) yield 0. □

Lemma 33 (Diagonal). $\binom{n}{n}_q = 1$ for every $n \in \mathbb{N}$.

Proof. Induction on n . The case $n = 0$ is by definition. For the step,

$$\binom{n+1}{n+1}_q = \binom{n}{n+1}_q + q^{n-n} \binom{n}{n}_q = 0 + 1 \cdot 1 = 1,$$

using Lemma 32 for the first term and the inductive hypothesis for the second. □

Lemma 34 (Closed form for $\binom{n}{k}_q$). *For all $k \leq n$,*

$$\binom{n}{k}_q \cdot (q; q)_k \cdot (q; q)_{n-k} = (q; q)_n.$$

Proof. Induction on n .

Base case $n = 0$. Forces $k = 0$ and the identity is $1 \cdot 1 \cdot 1 = 1$.

Boundary $k = 0$. Reduces to $1 \cdot 1 \cdot (q; q)_n = (q; q)_n$.

Boundary $k = n + 1$. By Lemma 33 we have $\binom{n+1}{n+1}_q = 1$, and $(q; q)_0 = 1$, so both sides equal $(q; q)_{n+1}$.

Inductive step $k + 1 \leq n$. Apply the q -Pascal recurrence

$$\binom{n+1}{k+1}_q = \binom{n}{k+1}_q + q^{n-k} \binom{n}{k}_q$$

and the factorisations

$$(q; q)_{n-k} = (q; q)_{n-(k+1)} \cdot (1 - q^{n-(k+1)}), \quad (q; q)_{k+1} = (q; q)_k \cdot (1 - q^{k+1}),$$

together with the power identities $q \cdot q^{n-(k+1)} = q^{n-k}$ and $q^{n-k} \cdot (q \cdot q^k) = q \cdot q^n$, which use $k \leq n$ via `Nat.sub_add_cancel`. The two inductive hypotheses at $(n, k+1)$ and (n, k) then combine into the required identity, dispatched in Lean by `linear_combination`. $\square$

6.3 The finite q -binomial theorem (step 5)

Theorem 25 is formalised as `qSeries.qBinom_finite_thm` in `Series/qSeries.lean`. The Lean proof follows exactly the four-step strategy spelled out in the proof attached to Theorem 25 above: peel off $k = 0$, apply the q -Pascal recurrence (Lemma 27) inside the sum, simplify the second piece to $zq^n Q_n$, and recombine the first piece via $\binom{n}{n+1}_q = 0$ (Lemma 32).

6.4 The infinite q -Pochhammer symbol and Cauchy identity (steps 6, 8)

Working in $\mathbb{C}$ for the analytic part of the development.

Definition 35 (Infinite q -Pochhammer symbol).

$$(a; q)_\infty := \prod_{k=0}^{\infty} (1 - a q^k) \in \mathbb{C},$$

defined unconditionally via `tprod`. The value is mathematically meaningful (and the product is convergent) under the hypothesis $\|q\| < 1$ provided by Lemma 36.

Lemma 36 (Multipliability). *For $a, q \in \mathbb{C}$ with $\|q\| < 1$, the family $k \mapsto 1 - a q^k$ is multipliable; in particular the partial products converge to $(a; q)_\infty$.*

Proof. Reduce multipliability to summability of $\sum_k \|a q^k\|$ via Mathlib's `multipliable_one_add_of_summable` (which works in any complete normed ring), then bound $\|a q^k\| = \|a\| \|q\|^k$ and conclude by the geometric series $\sum_k \|q\|^k < \infty$ (`summable_geometric_of_lt_one`). $\square$

Theorem 37 (Infinite q -binomial / Cauchy identity, restated for the formal development). *For $a, z, q \in \mathbb{C}$ with $\|q\| < 1$ and $\|z\| < 1$,*

$$\sum_{n=0}^{\infty} \frac{(a; q)_n}{(q; q)_n} z^n = \frac{(az; q)_\infty}{(z; q)_\infty}.$$

(Equivalent to Theorem 26.) The Lean theorem is `qSeries.qBinom_infinite_thm` in `Series/qSeries.lean`. The proof realised in Lean follows Heine’s functional-equation argument spelled out in the proof of Theorem 26 above, organised under **section CauchyIdentity**: non-vanishing of $(z; q)_\infty$ (`qPochhammerInf_z_q_ne_zero`), the telescoping recursion $(z; q)_\infty = (1-z)(zq; q)_\infty$ (`qPochhammerInf_recursion`), the coefficient recurrence $c_{n+1}(1 - q^{n+1}) = c_n(1 - aq^n)$ (`cauchyCoeff_succ_mul`), the functional equations for F and G (`cauchy_functional_eq_F`, `cauchy_functional_eq_G`), iteration on the disc (`iterated_functional_eq_disc`), and the Tannery-based limit $F(q^n z) \rightarrow 1$ (`tendsto_F_qpow`). The alternative route via a finite Cauchy-type analog of Theorem 25 is not formalised.

Remark. Together with Lemma 30, Lemma 32, and Lemma 34, Theorem 25 realises steps 1, 2, 3, 4, and 5 of the formalisation roadmap in Section 5 (“Formalization plan in Lean”). Step 6 is realised by Definition 35 and Lemma 36. Step 8 (the full Cauchy identity, Theorem 26) is now formalised in full via Heine’s functional-equation route (`qSeries.qBinom_infinite_thm`). Step 3 is partially completed: the recursive definition is in place (Definition 31) and the closed-form bridge to the quotient description is established (Lemma 34); the symmetry identity $\binom{n}{k}_q = \binom{n}{n-k}_q$ remains as future work, as does step 7 (the first Euler identity).

Update. Steps 7 and 8 are now fully formalised, and the Jacobi triple product and Euler’s pentagonal number theorem are proved with no sorry. The route actually taken by the formalisation diverges from all three textbook sketches above; it is documented in Section 7.

7 Implemented proof strategy

This section records the proof strategy that was *actually* carried out in the Lean formalisation (produced with the Aristotle prover). It does not follow any one of the three textbook routes of Sections 3.1, 4, 4.1 verbatim. It is closest in spirit to Proof 2 (the Euler / q -binomial route, Section 4), but the execution is neither Andrews’ nor Zhu’s: instead of Zhu’s semi-finite truncation or a complex-analytic continuation, the formalisation

1. expands $(q; q)_\infty(-z; q)_\infty$ and $(-q/z; q)_\infty$ by the *second* Euler identity and forms their full Cauchy product on $\mathbb{N} \times \mathbb{N}$;
2. evaluates every diagonal coefficient through a *novel key identity* $S_k(q) = 1/(q; q)_\infty$, proved by a contraction recurrence on the differences $S_k - S_{k+1}$ rather than by any combinatorial closed form;
3. removes the convergence (annulus) restriction by proving an *extended* second Euler identity valid for *all* $z \in \mathbb{C}$, so the Cauchy-product argument runs directly on the punctured disc; and
4. derives Euler’s pentagonal number theorem from the triple product by the substitution $q \mapsto q^3$, $z \mapsto -q$.

Source files. The development lives in the Lean library `qSeries_aristotle`, split into the modules `QSeries/Defs.lean`, `QSeries/FiniteBinomial.lean`, `QSeries/InfPochhammer.lean`, `QSeries/CauchyIdentity.lean`, `QSeries/EulerIdentities.lean`, `QSeries/JTP_KeyIdentity.lean`, `QSeries/JTP_Core.lean`, `QSeries/JacobiTripleProduct.lean`, and `QSeries/PentagonalNumber.lean`, all under the namespace `qSeries`. The q -Pochhammer / q -binomial foundations (Section 6) are shared with the monolithic `Series/qSeries.lean`; everything below builds on them.

7.1 The two Euler identities

Both Euler q -exponential identities are obtained as specialisations of the Cauchy identity (Theorem 37) and the finite q -binomial theorem (Theorem 25).

Lemma 38 (First Euler identity). *For $\|q\| < 1$ and $\|z\| < 1$,*

$$\sum_{n=0}^{\infty} \frac{z^n}{(q; q)_n} = \frac{1}{(z; q)_{\infty}}.$$

Proof. Specialise the Cauchy identity (Theorem 37) at $a = 0$, using $(0; q)_n = 1$ and $(0 \cdot z; q)_{\infty} = 1$. $\square$

Lemma 39 (Second Euler identity). *For $\|q\| < 1$ and $\|z\| < 1$,*

$$\sum_{n=0}^{\infty} \frac{q^{\binom{n}{2}} z^n}{(q; q)_n} = (-z; q)_{\infty}.$$

Proof. Take $N \rightarrow \infty$ in the finite q -binomial theorem (Theorem 25), rewriting $\prod_{k=0}^{N-1} (1 + zq^k) \rightarrow (-z; q)_{\infty}$ by Lemma 36 and replacing $\binom{N}{k}_q$ by $(q; q)_N / ((q; q)_k (q; q)_{N-k})$ (Lemma 34). The exchange of limit and sum is `tendsto_tsum_of_dominated_convergence`, with the dominating bound coming from $(q; q)_N / (q; q)_{N-k} \rightarrow 1$ being bounded uniformly in N . Summability of the target series is `qSeries.euler_second_summable`. $\square$

7.2 Expanding the product as a single series

The two factors $(q; q)_{\infty}(-z; q)_{\infty}$ are folded into one series using a telescoping identity for the infinite q -Pochhammer symbol.

Lemma 40 (Telescoping). *For $\|q\| < 1$ and every n ,*

$$(q; q)_{\infty} = (q; q)_n \cdot (q^{n+1}; q)_{\infty}.$$

Proof. Induction on n , using $(q; q)_{n+1} = (q; q)_n(1 - q^{n+1})$ (Lemma 30) and the recursion $(z; q)_{\infty} = (1 - z)(zq; q)_{\infty}$ (`qSeries.qPochhammerInf_recursion`). $\square$

Lemma 41 (Product as a series). *For $\|q\| < 1$ and $\|z\| < 1$,*

$$\sum_{n=0}^{\infty} q^{\binom{n}{2}} z^n (q^{n+1}; q)_{\infty} = (q; q)_{\infty} (-z; q)_{\infty}.$$

Proof. Multiply the second Euler identity (Lemma 39) by the constant $(q; q)_{\infty}$ and rewrite $(q; q)_{\infty} / (q; q)_n = (q^{n+1}; q)_{\infty}$ via Lemma 40. $\square$

7.3 The key identity $S_k = 1/(q; q)_{\infty}$

This is the novel core of the formalisation: it replaces both the coefficient-recurrence bookkeeping of Proof 1 and the complex analysis of Proof 3. After the Cauchy product is rearranged by diagonals, each diagonal coefficient is exactly one of the sums

$$S_k(q) := \sum_{m=0}^{\infty} \frac{q^{m(m+k)}}{(q; q)_m (q; q)_{m+k}},$$

and all of them collapse to $1/(q; q)_{\infty}$.

Definition 42 (The key sum). For $\|q\| < 1$ and $k \in \mathbb{N}$, $S_k(q) := \sum_{m \geq 0} q^{m(m+k)} / ((q; q)_m (q; q)_{m+k})$.

Lemma 43 (Summability of S_k). For $\|q\| < 1$, the summand of S_k is summable.

Proof. The denominators $(q; q)_m (q; q)_{m+k}$ are bounded below in norm by a positive constant (the partial products of a convergent infinite product with nonzero limit), so the summand is dominated by $C \|q\|^{m^2}$, which is summable. $\square$

Lemma 44 (Limit of S_k). For $\|q\| < 1$, $S_k(q) \rightarrow 1/(q; q)_\infty$ as $k \rightarrow \infty$.

Proof. The $m = 0$ term is $1/(q; q)_k \rightarrow 1/(q; q)_\infty$ (`tendsto_qPochhammer` and non-vanishing of $(q; q)_\infty$). The tail $m \geq 1$ tends to 0 by dominated convergence: each term $q^{m(m+k)} / ((q; q)_m (q; q)_{m+k}) \rightarrow 0$ as $k \rightarrow \infty$, dominated by $\|q\|^{m^2}/C$. $\square$

Lemma 45 (Recurrence for S_k). For $\|q\| < 1$ and every k ,

$$S_k - S_{k+1} = q^{k+1} (S_{k+2} - S_{k+1}).$$

Proof. Combine the two series term-by-term, using $(q; q)_{m+k+1} = (q; q)_{m+k} (1 - q^{m+k+1})$ to write $S_k - S_{k+1}$ as a single series, then split the numerator $1 - q^m - q^{m+k+1}$ and reindex the $(1 - q^m)$ part by $m \mapsto m + 1$. The two pieces reassemble into $q^{k+1} (S_{k+2} - S_{k+1})$. $\square$

Theorem 46 (Key identity). For $\|q\| < 1$ and every k , $S_k(q) = 1/(q; q)_\infty$.

Proof. Write $D_k = S_k - S_{k+1}$. Iterating Lemma 45 n times gives $\|D_k\| \leq \|q\|^{n(2k+n+1)/2} \|D_{k+n}\|$. Since $D_{k+n} \rightarrow 0$ (Lemma 44) and $\|q\|^n \rightarrow 0$, we get $D_k = 0$ for all k . Hence $S_k = S_0$ is constant in k , and its common value equals $\lim_k S_k = 1/(q; q)_\infty$. $\square$

Lemma 47 (Non-negative diagonal coefficient). For $\|q\| < 1$ and $k \geq 0$,

$$\sum_{m=0}^{\infty} q^{\binom{m+k}{2}} (q^{m+k+1}; q)_\infty \cdot \frac{q^{\binom{m}{2}} q^m}{(q; q)_m} = q^{\binom{k}{2}}.$$

Proof. Rewrite $(q^{m+k+1}; q)_\infty = (q; q)_\infty / (q; q)_{m+k}$ (Lemma 40) and use the arithmetic identity $\binom{m+k}{2} + \binom{m}{2} + m = \binom{k}{2} + m(m+k)$ to reduce the sum to $q^{\binom{k}{2}} (q; q)_\infty \cdot S_k(q)$. Apply Theorem 46. $\square$

Lemma 48 (Negative diagonal coefficient). For $\|q\| < 1$ and $l \geq 0$,

$$\sum_{n=0}^{\infty} q^{\binom{n}{2}} (q^{n+1}; q)_\infty \cdot \frac{q^{\binom{n+l+1}{2}} q^{n+l+1}}{(q; q)_{n+l+1}} = q^{\binom{l+2}{2}}.$$

Proof. As in Lemma 47, using the companion arithmetic identity $\binom{n}{2} + \binom{n+l+1}{2} + (n+l+1) = \binom{l+2}{2} + n(n+l+1)$ and the key identity $S_{l+1} = 1/(q; q)_\infty$ (Theorem 46). $\square$

7.4 Extended second Euler identity (all z)

To run the Cauchy product on the full punctured disc rather than only the annulus, the second Euler identity is extended to all $z \in \mathbb{C}$.

Lemma 49 (Series recursion). For $\|q\| < 1$, the series $S(z) := \sum_n q^{\binom{n}{2}} z^n / (q; q)_n$ satisfies $S(z) = (1+z) S(qz)$.

Proof. A term-by-term algebraic identity: the difference of the z - and qz -series equals $zS(qz)$, using $(q; q)_n = (q; q)_{n-1}(1-q^n)$. Summability for all z is `qSeries.euler_second_summable_all`, proved by the ratio test (the ratio $\|q\|^n \|z\| / \|1 - q^{n+1}\| \rightarrow 0$). $\square$

Theorem 50 (Extended second Euler identity). *For $\|q\| < 1$ and every $z \in \mathbb{C}$,*

$$\sum_{n=0}^{\infty} \frac{q^{\binom{n}{2}} z^n}{(q; q)_n} = (-z; q)_{\infty}.$$

Proof. Given any z , choose N with $\|zq^N\| < 1$. Iterating Lemma 49 N times gives $S(z) = \prod_{k=0}^{N-1} (1 + zq^k) \cdot S(zq^N)$, and $S(zq^N) = (-zq^N; q)_{\infty}$ by the (small-argument) second Euler identity (Lemma 39). The product $\prod_{k=0}^{N-1} (1 + zq^k) \cdot (-zq^N; q)_{\infty} = (-z; q)_{\infty}$ by repeated use of `qSeries.qPochhammerInf_recursion`. $\square$

Lemma 51 (Extended Euler identity at q/z). *For $\|q\| < 1$ and $z \neq 0$ (no annulus condition),*

$$\sum_{m=0}^{\infty} \frac{q^{\binom{m}{2}} q^m z^{-m}}{(q; q)_m} = (-q/z; q)_{\infty}.$$

Proof. Apply Theorem 50 at the point q/z (valid for all q/z , hence all $z \neq 0$). $\square$

7.5 The triple product and bilateral series

Definition 52 (Triple product and bilateral series). $\text{jacobiProd}(q, z) := (q; q)_{\infty} (-z; q)_{\infty} (-q/z; q)_{\infty}$, and the bilateral series $\text{jacobiBilateral}(q, z) := \sum_{k \geq 0} z^k q^{\binom{k}{2}} + \sum_{m \geq 0} z^{-(m+1)} q^{\binom{m+2}{2}}$ (the non-negative and negative halves of $\sum_{k \in \mathbb{Z}} z^k q^{k(k-1)/2}$).

7.6 The Jacobi triple product

Theorem 53 (Jacobi triple product). *For $\|q\| < 1$, $\|z\| < 1$, $z \neq 0$, $\text{jacobiProd}(q, z) = \text{jacobiBilateral}(q, z)$; that is,*

$$(q; q)_{\infty} (-z; q)_{\infty} (-q/z; q)_{\infty} = \sum_{k \geq 0} z^k q^{\binom{k}{2}} + \sum_{m \geq 0} z^{-(m+1)} q^{\binom{m+2}{2}}.$$

Proof. Expand $(q; q)_{\infty} (-z; q)_{\infty}$ as a series (Lemma 41) and $(-q/z; q)_{\infty}$ via the *extended* second Euler identity (Lemma 51), which holds for every $z \neq 0$ with no lower bound on $\|z\|$. Their Cauchy product over $\mathbb{N} \times \mathbb{N}$ is summable (`Summable.mul_norm`), so by `Summable.tsum_prod` the double sum may be reordered. Partition $\mathbb{N} \times \mathbb{N}$ into the diagonals $n \geq m$ (index $k = n - m \geq 0$) and $n < m$ (index $l = m - n - 1 \geq 0$). The k -diagonal sum is $z^k q^{\binom{k}{2}}$ by Lemma 47, and the l -diagonal sum is $z^{-(l+1)} q^{\binom{l+2}{2}}$ by Lemma 48. Since the extended identity carries no annulus hypothesis, the argument runs directly on the full punctured disc $0 < \|z\| < 1$, with no separate continuation step. $\square$

Remark (divergence from the outline). Compared with Section 4: the formalisation uses a *genuine Cauchy product* of two second-Euler expansions, not Zhu’s semi-finite truncation; the diagonal coefficients are evaluated through the key identity Theorem 46, whose contraction-recurrence proof has no analogue in any of the three sketches; and the argument is carried out directly on the whole punctured disc by strengthening the Euler identity to all z (Theorem 50), so it needs neither an annulus restriction with a separate extension step, the Liouville-type complex analysis of Proof 3, nor the coefficient bookkeeping of Proof 1.

7.7 Euler's pentagonal number theorem

Definition 54 (Generalized pentagonal number). $\omega(k) := (k(3k-1))/2$ for $k \in \mathbb{Z}$ (the product $k(3k-1)$ is always even), realised in Lean as a natural number.

Theorem 55 (Euler's pentagonal number theorem). *For $\|q\| < 1$,*

$$(q; q)_{\infty} = \sum_{k \geq 0} (-1)^k q^{\omega(k)} + \sum_{k \geq 0} (-1)^{k+1} q^{\omega(-(k+1))} = 1 - q - q^2 + q^5 + q^7 - q^{12} - \dots$$

Proof. Apply the Jacobi triple product (Theorem 53) with $q \mapsto q^3$ and $z \mapsto -q$; the hypotheses reduce to $\|-q\| < 1$ and $-q \neq 0$, both immediate for $0 < \|q\| < 1$ (the case $q = 0$ is trivial). The left-hand side $(q^3; q^3)_{\infty} (q; q^3)_{\infty} (q^2; q^3)_{\infty}$ is regrouped into $(q; q)_{\infty}$ by splitting the product $\prod_{n \geq 1} (1 - q^n)$ into the three residue classes $\{3k+1\}, \{3k+2\}, \{3k+3\}$ modulo 3. Matching the right-hand exponents k^2 (with sign $(-1)^k$) against $\omega(\pm k)$ gives the pentagonal series. $\square$